



Mobile Application System Design

(Chapter 5)

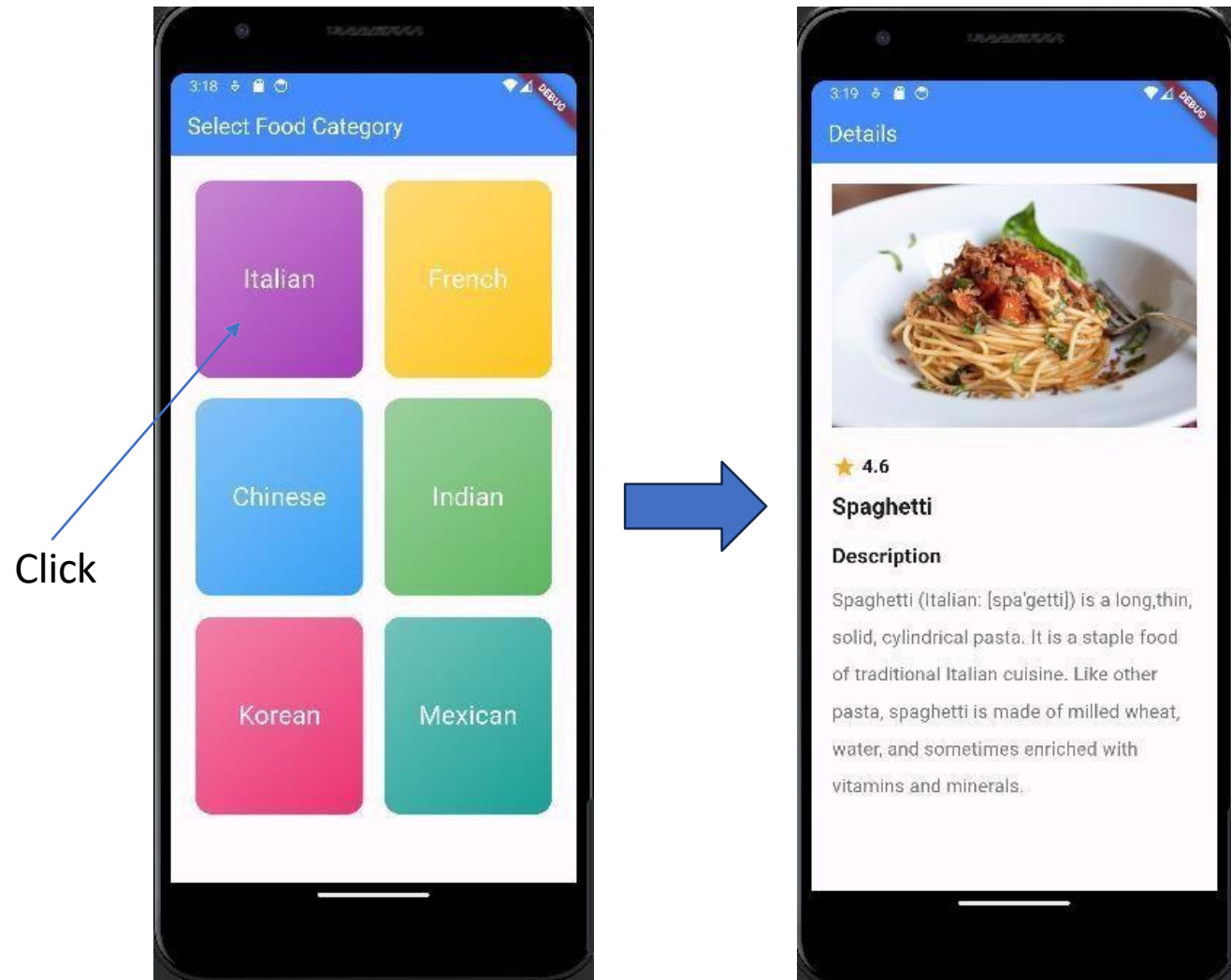
Course Lecturer: Nurul Amera Muhamad Nazmi

CLASS ACTIVITY

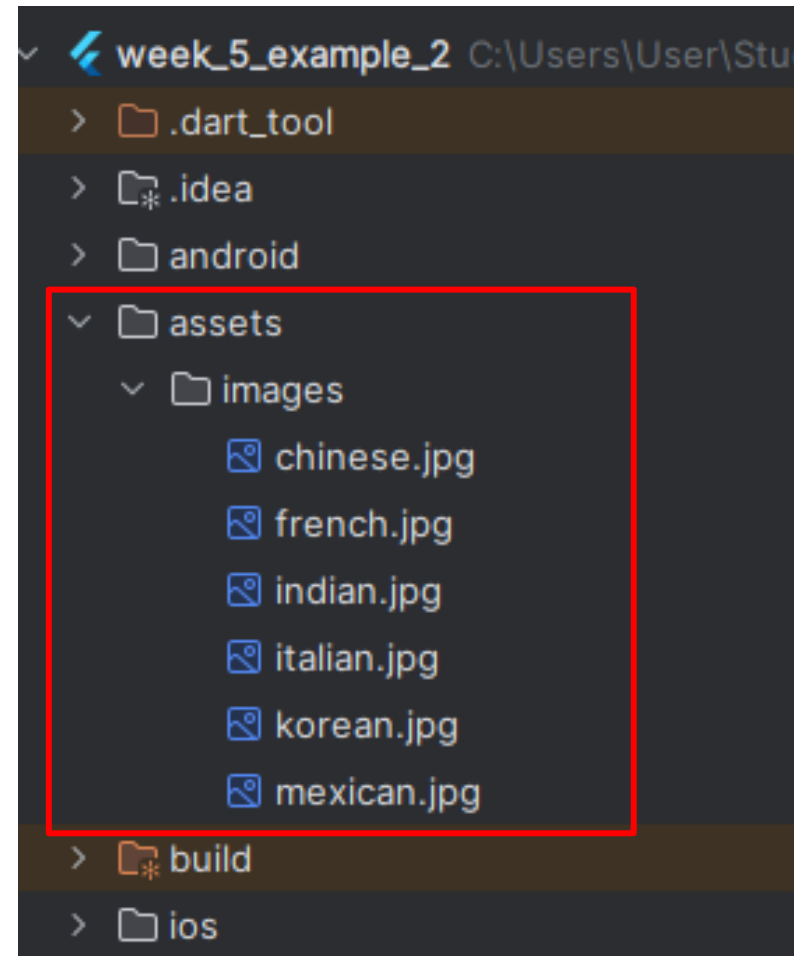
(Hands-On Practice)

Food Category App

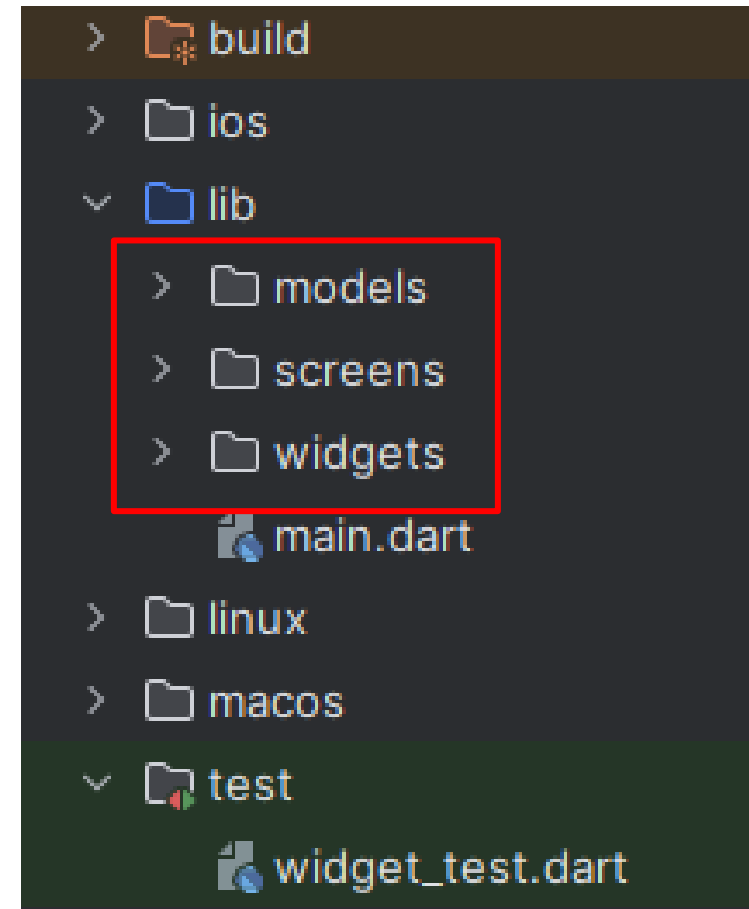
- In this example, we will create a menu page using GridView widget.



- Create a new Flutter Project.
- Create a folder and a sub-folder named “assets” and “images”, respectively.
- Copy and paste the resources images downloaded from Moodle to the images folder.
- Set the image path in pubspec.yaml.



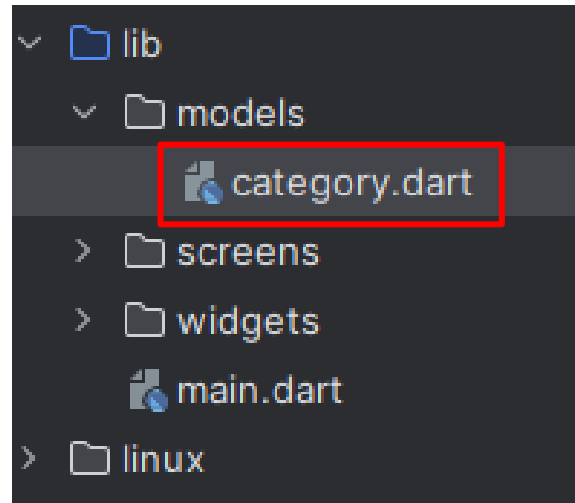
- Create three subfolders named **models**, **screens**, and **widgets**, inside the lib folder.



- In the main.dart file, create a main function to kickstart the app running.

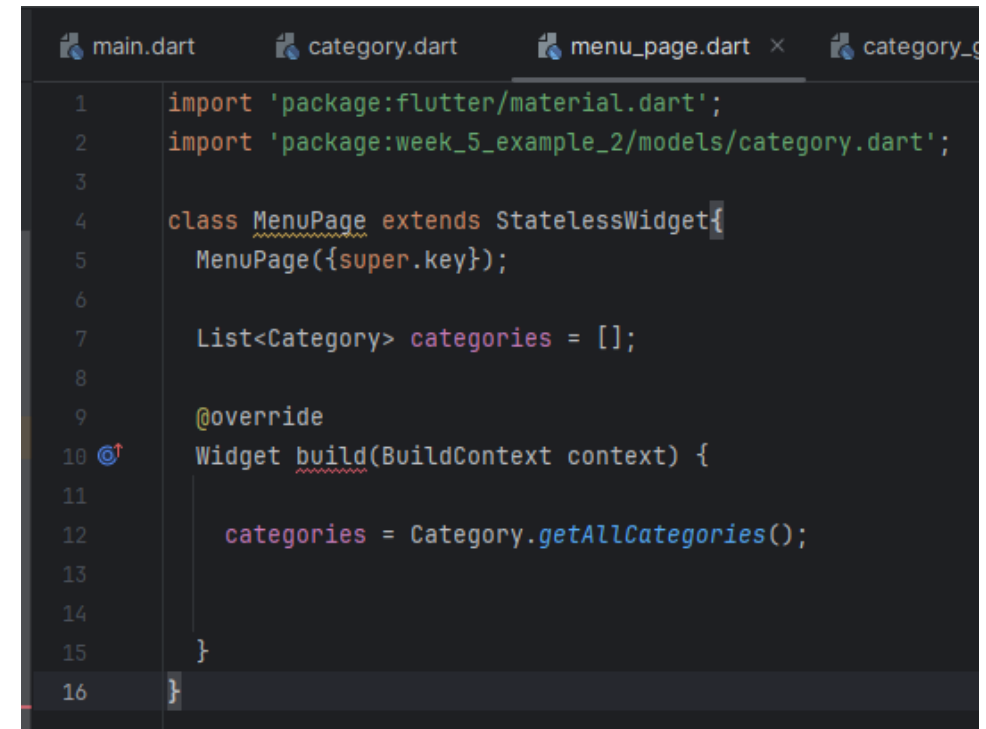
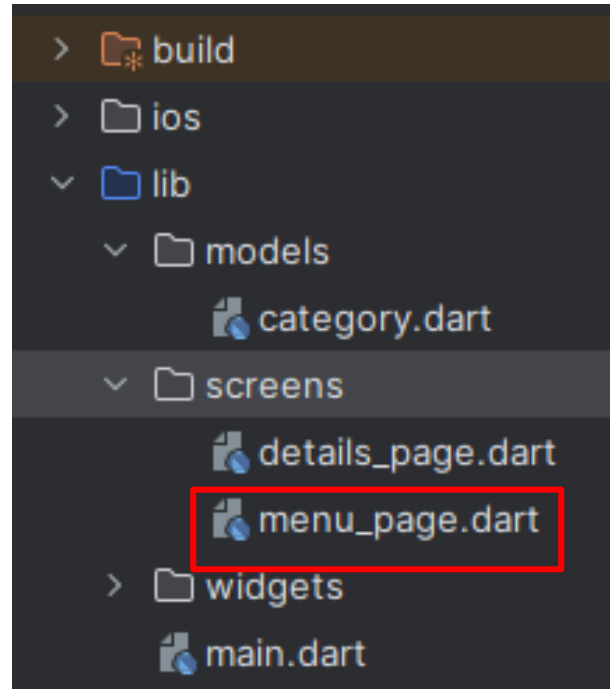
```
main.dart x
1  import 'package:flutter/material.dart';
2  import 'package:week_5_example_2/screens/menu_page.dart';
3
4
5  >> void main() {
6      runApp(
7          MaterialApp(
8              title: 'Food Intro App',
9              home: MenuPage()
10             ) // MaterialApp
11      );
12  }
```

- Create a new dart file named “category.dart” file in the models folder.
- Copy and paste the scripts from category.txt into the category.dart file.



```
main.dart  category_grid_item.dart  category.dart  details_page.dart  menu_page.dart
lib > models > category.dart > Category > getAllCategories
1  import 'package:flutter/material.dart';
2
3  class Category {
4    String id;
5    String title;
6    Color color;
7    String imagePath;
8    double rating;
9    String foodName;
10   String description;
11
12   Category({
13     required this.id,
14     required this.title,
15     required this.color,
16     required this.imagePath,
17     required this.rating,
18     required this.foodName,
19     required this.description,
20   });
21
22   // Sample category data
23   static List<Category> getAllCategories() {
24     List<Category> categories = [];
25
26     categories.add(
27       Category(
28         id: 'c1',
29         title: 'Italian',
30         color: Colors.purple,
31         imagePath: 'assets/images/italian.jpg',
32         rating: 4.5,
33         foodName: 'Lasagna',
34         description: 'Lasagna is a traditional Italian dish made by stacking layers '
35           'of pasta sheets with rich meat or vegetable sauce, creamy béchamel, '
36           'and cheese. It is then baked to create a flavorful and comforting meal.',
37       ),
38     );
39   }
```

- Create a new dart file named “**menu_page.dart**” file in the screens folder.
- In **menu_page.dart** file, create a MenuPage class that extends the StatelessWidget.
- Create a categories list and inside the build function, invokes the getAllCategories() function to extract all categories info from the data class model.

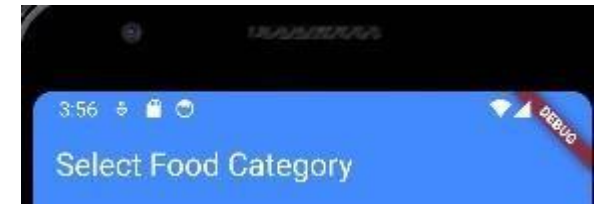


- Add a Scaffold widget to the MenuPage.
- Create an AppBar by setting a title.
- Set the font in white color and the background as blue accent.

```
@override
Widget build(BuildContext context) {

  categories = Category.getAllCategories();

  return Scaffold(
    appBar: AppBar(
      title: const Text(
        "Select Food Category",
        style: TextStyle(
          color: Colors.white,
        ) // TextStyle
      ), // Text
      backgroundColor: Colors.blueAccent,
    ), // AppBar
  ); // Scaffold
}
```



- Add a Scaffold widget to the MenuPage.
- Create an AppBar by setting a title.
- Set the font in white color and the background as blue accent.

```
return Scaffold(  
  appBar: AppBar(  
    title: const Text(  
      "Select Food Category",  
      style: TextStyle(  
        color: Colors.white,  
      ) // TextStyle  
    ), // Text  
    backgroundColor: Colors.blueAccent,  
  ), // AppBar  
  body: GridView(  
    padding: const EdgeInsets.all(24),  
    gridDelegate: const SliverGridDelegateWithFixedCrossAxisCount(  
      crossAxisCount: 2,  
      childAspectRatio: 8.5/10,  
      crossAxisSpacing: 20,  
      mainAxisSpacing: 20  
    ), // SliverGridDelegateWithFixedCrossAxisCount  
    children:[  
  
    ],  
  ), // GridView  
); // Scaffold
```

```

class MenuPage extends StatelessWidget {
  MenuPage({super.key});

  List<Category> categories = [];

  @override
  Widget build(BuildContext context) {
    categories = Category.getAllCategories();

    void selectFoodCategory(context, Category category)
    {
      Navigator.of(context).push(
        MaterialPageRoute(
          builder: (ctx)=> DetailsPage(
            category : category,
          ) // DetailsPage
        ) // MaterialPageRoute
      );
    }

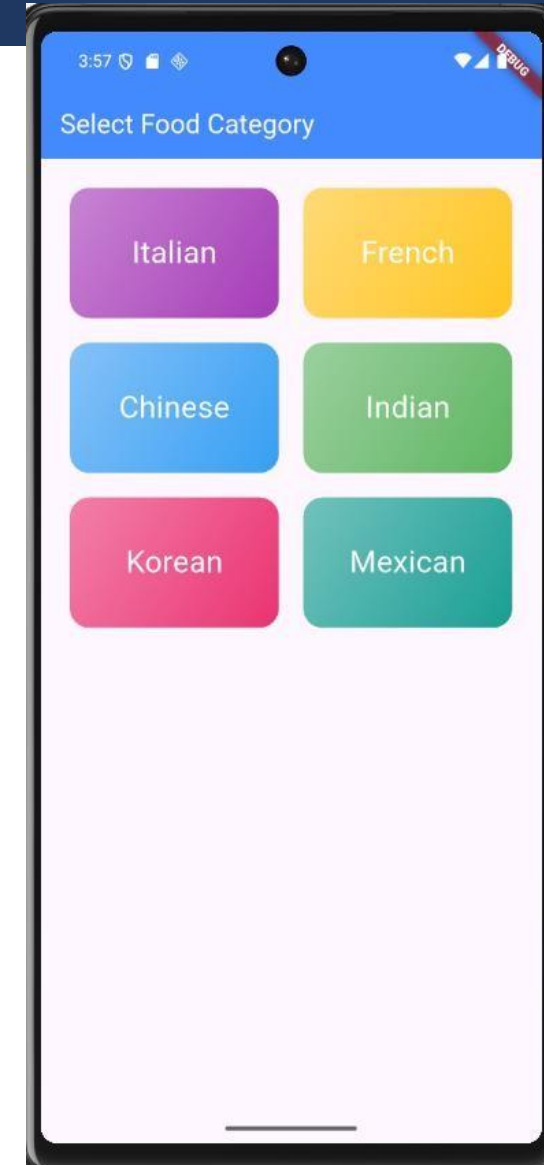
    return Scaffold(
      appBar: AppBar(
        title: const Text(
          'Select Food Category',
          style: TextStyle(color: Colors.white),
        ), // Text
        backgroundColor: Colors.blueAccent,
      ), // AppBar
      body: GridView(
        padding: const EdgeInsets.all(24),
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          crossAxisCount: 2,
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```

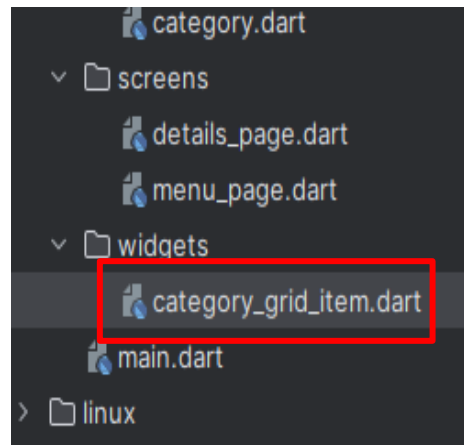
- In the menu_page.dart file, add an import statement to import the category_grid_item.dart.
- Create a for loop to generate the item in the GridView

```
import 'package:flutter/material.dart';  
import 'package:chap5_example2/models/category.dart';  
import 'package:chap5_example2/widgets/category_grid_item.dart';  
import 'package:chap5_example2/screens/details_page.dart';
```

```
gridDelegate: const SliverGridDelegateWithFixedCrossAxisCount(  
  crossAxisCount: 2,  
  childAspectRatio: 8 / 5,  
  crossAxisSpacing: 20,  
  mainAxisSpacing: 20,  
), // SliverGridDelegateWithFixedCrossAxisCount  
// Step 2: Generate children using a for loop  
children: [  
  for (final category in categories)  
    CategoryGridItem(  
      category: category,  
      onSelectCategory: () {  
        selectFoodCategory(context, category);  
      },  
    ), // CategoryGridItem  
],  
), // GridView  
); // Scaffold  
}
```



- Add a new dart file named 'category_grid_item.dart' inside the widgets folder.
- Copy and paste the scripts from 'category_grid_item.txt' given in the resource folder to the category_grid_item.dart.



```
import 'package:flutter/material.dart';
import 'package:chap5_example2/models/category.dart';

class CategoryGridItem extends StatelessWidget {
  const CategoryGridItem({
    super.key,
    required this.category,
    required this.onSelectCategory,
  });

  final Category category;
  final void Function() onSelectCategory;

  @override
  Widget build(BuildContext context) {
    return InkWell(
      onTap: onSelectCategory, // updated to use function from parent
      splashColor: Theme.of(context).primaryColor,
      borderRadius: BorderRadius.circular(16), // match snapshot
      child: Container(
        padding: const EdgeInsets.all(16),
        alignment: Alignment.center,
        decoration: BoxDecoration(
          borderRadius: BorderRadius.circular(16),
          gradient: LinearGradient(
            colors: [
              category.color.withOpacity(0.55), // softer shade
              category.color.withOpacity(0.9), // stronger shade
            ],
            begin: Alignment.topLeft,
            end: Alignment.bottomRight,
          ), // LinearGradient
        ), // BoxDecoration
        child: Text(
          category.title,
          style: const TextStyle(
            color: Colors.white,
            fontSize: 25,
          ), // TextStyle
        ), // Text
      ), // Container
    ); // InkWell
  }
}
```

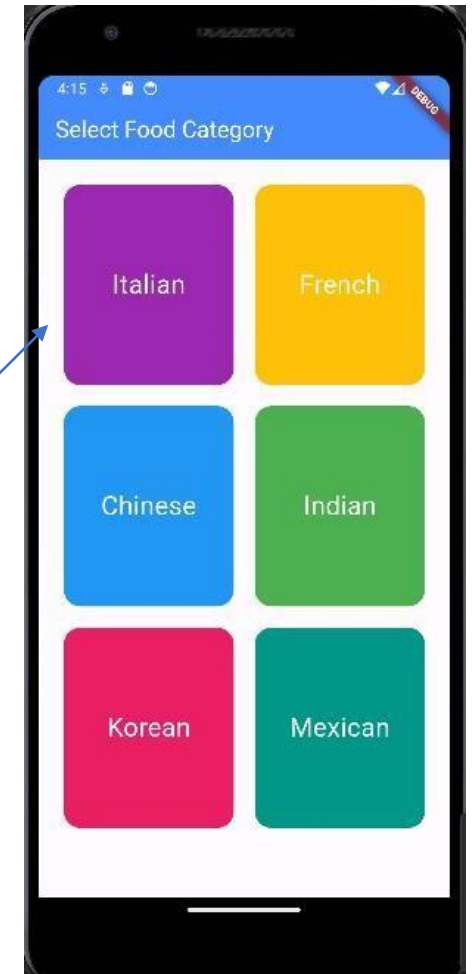
```
import 'package:flutter/material.dart';
import 'package:chap5_example2/models/category.dart';

class CategoryGridItem extends StatelessWidget {
  const CategoryGridItem({
    super.key,
    required this.category,
    required this.onSelectCategory,
  });

  final Category category;
  final void Function() onSelectCategory;

  @override
  Widget build(BuildContext context) {
    return InkWell(
      onTap: onSelectCategory, // updated to use function from parent
      splashColor: Theme.of(context).primaryColor,
      borderRadius: BorderRadius.circular(16), // match snapshot
      child: Container(
        padding: const EdgeInsets.all(16),
        alignment: Alignment.center,
        decoration: BoxDecoration(
          borderRadius: BorderRadius.circular(16),
          gradient: LinearGradient(
            colors: [
              category.color.withOpacity(0.55), // softer shade
              category.color.withOpacity(0.9), // stronger shade
            ],
            begin: Alignment.topLeft,
            end: Alignment.bottomRight,
          ), // LinearGradient
        ), // BoxDecoration
      ), // InkWell
    );
  }
}
```

A widget that effectively simulates material design ink splash on any widget for visual interactivity



See you next class